Exercise 2 of module 1004D on "Theoretical spectroscopy for systems implying periodic boundary conditions"

Functional derivatives

a The functional derivative of a function $F[\rho] = \int f(\rho(\mathbf{r}))\mathrm{d}^3 r$ is given as

$$\frac{\delta F[\rho]}{\delta \rho} = \frac{\partial f}{\partial \rho}\,. \tag{1}$$

Derive this formula using the definition of the functional derivative,

$$F[f+\delta f] = F[f] + \int \frac{\delta F[f]}{\delta f(x)}\delta f(x)\mathrm{d}r + \mathcal{O}[(\delta f)^2]\,, \tag{2}$$

by expanding $f[\rho]$ as a Taylor series around $\rho = \rho(\mathbf{r})$.

b Using the definition of a functional derivative, derive the effective Kohn-Sham potential within the local density approximation,

$$v_{\text{eff}}^{\text{LDAx}}(\mathbf{r}) = \frac{\delta\left(J[\rho] + E_{\text{x}}^{\text{LDA}}[\rho] + V_{\text{nuc}}[\rho]\right)}{\delta\rho(\mathbf{r})}\,, \tag{3}$$

with

$$E_{\text{x}}^{\text{LDA}}[\rho] = -C_{\text{x}}\int \rho^{4/3}(\mathbf{r})\mathrm{d}^3 r\,, \tag{4}$$

$$V_{\text{nuc}}[\rho] = \int v_{\text{nuc}}(\mathbf{r})\rho(\mathbf{r})\mathrm{d}^3 r\,, \tag{5}$$

$$J[\rho] = \frac{1}{2}\int\int \frac{\rho(\mathbf{r}_1)\rho(\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}\mathrm{d}^3 r_1 \mathrm{d}^3 r_2\,. \tag{6}$$

c Compute the derivative of the Thomas-Fermi functional for the kinetic energy,

$$\frac{\delta T_{\text{TF}}[\rho]}{\delta\rho(\mathbf{r})}\,, \tag{7}$$

with

$$T_{\text{TF}}[\rho] = C_{\text{F}}\int \rho^{5/3}(\mathbf{r})\mathrm{d}^3 r\,. \tag{8}$$

d Show that the derivative of the von Weizsacker functional for the kinetic energy is given as

$$\frac{\delta T_{\text{vW}}[\rho,\nabla\rho]}{\delta\rho(\mathbf{r})} = -\frac{1}{4}\frac{\nabla^2\rho(\mathbf{r})}{\rho(\mathbf{r})} + \frac{1}{8}\frac{(\nabla\rho(\mathbf{r}))^2}{\rho(\mathbf{r})^2}\,, \tag{9}$$

with

$$T_{\text{vW}}[\rho,\nabla\rho] = \frac{1}{8}\int \frac{(\nabla\rho(\mathbf{r}))^2}{\rho(\mathbf{r})}d^3 r\,. \tag{10}$$

e Show that the funtional derivative of GGA functionals,

$$E_{\mathrm{xc}}^{\mathrm{GGA}}[\rho] = \int F(\rho(\mathbf{r}), \nabla\rho(\mathbf{r}))\mathrm{d}^3 r\,, \tag{11}$$

is given as

$$\frac{\delta E_{\mathrm{xc}}^{\mathrm{GGA}}[\rho]}{\delta\rho(\mathbf{r})} = \left.\frac{\partial F(\rho, \nabla\rho)}{\partial\rho}\right|_{\rho=\rho(\mathbf{r});\nabla\rho=\nabla\rho(\mathbf{r})} - \nabla\cdot\left.\frac{\partial F(\rho, \nabla\rho)}{\partial\nabla\rho}\right|_{\rho=\rho(\mathbf{r});\nabla\rho=\nabla\rho(\mathbf{r})}. \tag{12}$$

Reduced density matrices

The total energy can be expressed in terms of the density $\rho(\mathbf{r})$, the reduced one-electron density matrix $\gamma_1(\mathbf{x}, \mathbf{x}')$ and the electron pair density $\rho_2(\mathbf{r}_1, \mathbf{r}_2)$. All these quantities can be derived from the reduced two-particle density matrix $\gamma_2(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_1', \mathbf{x}_2')$,

$$\gamma_2(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_1', \mathbf{x}_2') = \frac{N(N-1)}{2}\int \Psi^*(\mathbf{x}_1', \mathbf{x}_2', \mathbf{x}_3, \ldots, \mathbf{x}_N)\Psi(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \ldots, \mathbf{x}_N)\mathrm{d}\mathbf{x}_3 \ldots \mathrm{d}\mathbf{x}_N\,. \tag{13}$$

a Compare the definition of $\gamma_2(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_1', \mathbf{x}_2')$ with the definitions for $\rho(\mathbf{r})$, $\gamma_1(\mathbf{x}, \mathbf{x}')$ and $\rho_2(\mathbf{r}_1, \mathbf{r}_2)$ (see lecture). How can $\rho(\mathbf{r})$, $\gamma_1(\mathbf{x}, \mathbf{x}')$ and $\rho_2(\mathbf{r}_1, \mathbf{r}_2)$ be computed from the reduced two-particle density matrix $\gamma_2(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_1', \mathbf{x}_2')$?

b Can the Pauli exclusion principle be derived from the reduced two-particle density matrix? (How can the probability of finding two electrons of the same spin at the same place be computed from the reduced two-particle density matrix? Why has this probability to be equal zero?)

Back to HF theory: Slater determinants, density and density matrices

a The electron density $\rho(\mathbf{r})$ can be expressed as an expectation value over the one-electron density operator,

$$\hat{\rho} = \sum_{i=1}^{N} \delta(\mathbf{r} - \mathbf{r}_i)\,. \tag{14}$$

Compute the electron density of a Slater determinant $|\Phi^{\mathrm{SD}}$.

b Show that the pair density $\rho_2(\mathbf{r}_1, \mathbf{r}_2)$ can be written as an expectation value of the operator

$$\hat{\rho}_2 = \sum_{i=1}^{N}\sum_{j=i+1}^{N} \delta(\mathbf{r}' - \mathbf{r}_i)\delta(\mathbf{r}'' - \mathbf{r}_j)\,. \tag{15}$$

c Use $\hat{\rho}_2$ to compute the pair density $\rho_2(\mathbf{r}_1, \mathbf{r}_2)$ of a Slater determinant.

d The reduced one-electron density matrix $\gamma_1(\mathbf{x}, \mathbf{x}')$ can not be straight-forwardly written as the expectation value of an operator. Compute $\gamma_1(\mathbf{x}, \mathbf{x}')$ of a Slater determinant.

e Show that Slater determinants fulfill the following relation:

$$\rho_2(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{2}[\rho(\mathbf{r}_1)\rho(\mathbf{r}_2) - |\rho_1(\mathbf{r}_1, \mathbf{r}_2)|^2]\,. \tag{16}$$

f Reformulate the Hartree-Fock energy in terms of $\rho(\mathbf{r})$ and $\rho_1(\mathbf{r}, \mathbf{r}')$.